Reputation is Amortized Verification:

Inspection Games for Agent Economies

Paper 3 of the Harbor research program — B2/R7, executed

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Abstract

A marketplace where AI judges grade AI work is an inspection game. We give three results. *Stage:* a bonded judge is deterred iff $\rho dB \geq G$; the critical audit rate is $\rho^\star = G/(dB)$ under the keep-gain convention and $\rho^\star_c = G/(d(G{+}B))$ under confiscation — the settlement convention alone moves the headline audit budget by roughly 17%. *Tower:* when level $k{+}1$ audits level-k auditors sampled *sealed* from a pool spanning C disjoint cliques, rational bribery is all-or-nothing and profitable only while corrupt value exceeds $C\,B$; below that threshold corruption decays geometrically at $(1-\rho d)$ per level, so *finite* bond capital certifies a tower deep enough to drive corrupt value below any fixed unit — a depth logarithmic in that value. *Amortization:* as a judge's verified history grows, incentive-compatible audit schedules decline — lifetime audit spend falls from $\Theta(T)$ (flat) to $\Theta(\log T)$ (losses only if audited) to $O(1)$, with the constant $aG^2/(2dBrv)$ (cheats also surface at an independent rate r). Reputation, on this account, is not a soft social layer bolted onto verification: it is the mechanism by which verification cost is amortized over a history. The results convert the source volume's central open conjecture (III.11.1) into a parameterized theorem with a named implementation, and every number reproduces from a bundled script.

Express lane: deterrence holds iff audit-probability $\times$ detection $\times$ bond $\geq$ corrupt gain; stacking auditors sampled sealed from C independent cliques shuts off the bribery-profitable phase, leaving a geometric collapse that finite total bond certifies at logarithmic depth; and history-indexed audit schedules make lifetime verification spend $\Theta(\log T)$ or $O(1)$ instead of $\Theta(T)$ — formal statements in the boxes of §4, §5, §6.

1 Introduction

The scene. A marketplace where AI agents do work and AI judges grade it. A judge posts a bond that is slashed if it is caught cheating. A judge can take a bribe: pass shoddy work, pocket G. So you audit the judges — but auditors can be bribed too, and now it looks like watchers all the way up, each level dearer than the last. Meanwhile the platform's accountant asks a different question: a judge with ten thousand verified grades is still being audited at the same rate as a newcomer — is that spend buying anything?

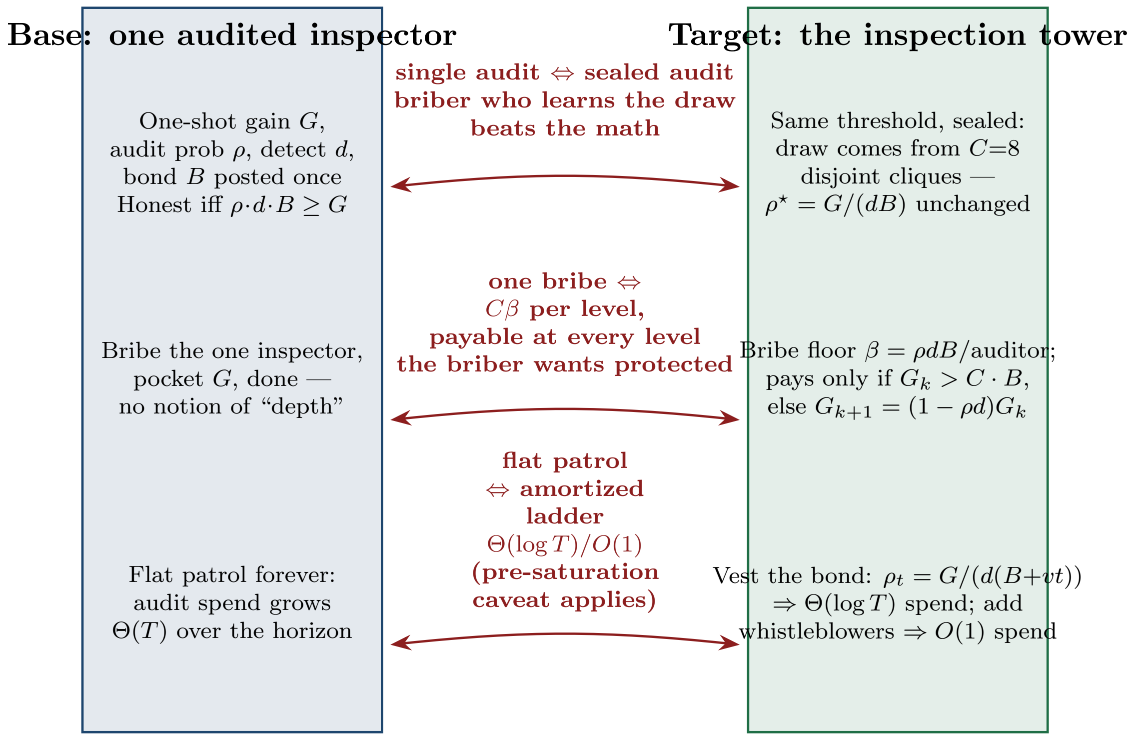
The idea in one breath.

Honesty holds when expected punishment beats the bribe; the tower of auditors converges on finite bond capital because each level has geometrically less corruption left to protect, and drawing judges sealed from C independent camps buys down the depth by shutting off the phase where bribery still pays; and the audit rate an old judge needs falls with the reputation it would forfeit — so reputation is amortized verification.

The idea, before the formalism. Traffic enforcement does not post an officer on every corner; it makes expected fine $\times$ odds of being caught exceed the minutes saved by speeding.

That is the stage game: gain G, audit probability ρ, detection probability d, slashable bond B. The tower adds jury selection from disjoint pools: if each auditing layer is drawn sealed from C rival benches, a briber must pay all C benches to be safe — worthwhile only for enormous corruption, which then bleeds value paying for its own protection. And the amortization result is the actuarial half of the same logic: a judge with a long verified history has a large future income stream at stake, which is itself a punishment the platform did not have to fund — so the platform can audit it less, and the incentive constraint stays exactly binding. Figure 1 draws the correspondence. *Misread to preempt:* the tower result does *not* require infinitely many honest auditors, and the amortization result does *not* assume old judges are intrinsically nicer. The former needs only finite bonds and a sealed draw, because what each level protects shrinks geometrically once bribery stops paying; clique diversity ($C > 1$) buys depth and capital, not convergence. The latter needs only that accumulated reputation is genuinely forfeit on capture — deterrence is re-derived, not assumed, at every t.

R7 — reputation is amortized verification



The tower prices depth, not detection: finite bond, sealed against the draw, certifies a depth logarithmic in the corrupt value — 27 levels at C=8, 53 at C=1.

Figure 1: The relation map: the stage game (left) and the tower (right) are the same deterrence arithmetic, applied once and applied recursively. The arrows carry the load — a single committed audit maps to a *sealed* draw from C benches; one bribe maps to a bribe bill of $C\beta$ that must be paid *at every level* the briber wants protected; and a flat patrol budget maps to a history-indexed schedule. The relation preserved across all three rows is *expected forfeiture* $\geq$ *corrupt gain*; what changes is only who pays it and over what horizon. What C buys is depth and capital (53 levels and 2650 at C=1 against 27 and 1350 at C=8), not the difference between a tower that terminates and one that does not.

Contributions. (i) The stage-game threshold with its convention fork made explicit (§4) — the same game yields $\rho^\star = 0.25$ or $\rho^\star_c = 0.2083$ at our running parameters depending on whether capture confiscates the gain, and quoting one without pinning the settlement rule is a reproducible way to be 17% wrong. (ii) The tower-contraction theorem (§5): sealed sampling from C disjoint cliques makes bribery uneconomical below corrupt value $C\,B$, after which corruption decays geometrically — so finite bond capital certifies an audit depth logarithmic in the corrupt value, with sealing identified as load-bearing rather than hygienic and the clique multiplier priced honestly (it removes the linear bribery phase; it is not what makes the tower converge). (iii) The amortization theorems (§6): flat auditing spends $\Theta(T)$; the loss-only-if-audited schedule spends $\Theta(\log T)$; independent revelation at rate r gives $O(1)$ lifetime spend, with the constant $aG^2/(2dBrv)$ — and which of the two holds is an empirical fork, not a modeling taste. (iv) The connection to the program's repeated-game mechanization (§7): the discount-factor thresholds $\delta \geq 1/3$ (grim) and $\delta^\star = 0.3425080314$ (graduated trigger), now CI-checked by Z3 and TLC/Apalache in the repository's `proofs/economics/` directory, are the same deterrence arithmetic with continuation value in place of bonds. (v) The conversion of the source volume's Conjecture III.11.1 into a parameterized theorem with a named implementation (§8). Every number is tagged [verified] (externally recomputable) or [internal] (regenerates from the bundled script `b2_tower.py`).

2 Related work, and what is actually new

The stage game is classical. Deterrence-by-expected-penalty is Becker [2]; the leader–follower and simultaneous inspection formulations, including the committed-inspector solution we use, are surveyed by Avenhaus, von Stengel and Zamir [1], whose treatment of inspector commitment and inspectee indifference we follow. Reputation-as-equilibrium-behavior is Kreps–Wilson [3] and Fudenberg–Levine [4]: a long-lived player's continuation value disciplines short-run temptation. Batched verification of many items with few tests goes back to Dorfman's group-screening design [5], which is the right lineage for pooled audit sampling, though we do not use pooling here.

New, honestly stated: the amortized-verification framing — deriving the *declining audit schedule* that keeps the incentive constraint exactly binding as history accumulates, and costing it end-to-end ($\Theta(T)$ vs. $\Theta(\log T)$ vs. $O(1)$, with the closed form $aG^2/(2dBrv)$); the contraction theorem for a *stackable* audit tower whose auditors are themselves corruptible LLM judges, with model heterogeneity identified as the mechanism that supplies the disjoint cliques the contraction needs; and the observation that sealing the sampling draw is load-bearing (§5). *Imported:* everything in the previous paragraph. The contribution is closing a conjecture the source volume posed, with a named implementation and a measurement obligation — not a new solution concept.

3 The vocabulary, defined

Express lane: skip to §4 — nothing in this section is needed to follow the theorems, only to read them comfortably; the two terms that are hypotheses rather than vocabulary, sealed *and* clique*, are defined where they first do work, at the head of §5.* This paper is economics, and its terms are cheap to define but expensive to guess at. A software reader who has never taken a mechanism-design course loses nothing but the dictionary; here it is, in the order the terms are needed.

An *inspection game* is a two-player game between someone who may cheat and someone who may check, where checking costs something and catching pays something. Its defining feature — and the reason it is not simply a matter of "audit more" — is that it typically has no equilibrium in which either player behaves predictably: if the inspector always checks, cheating never pays and the checks are wasted; if the inspector never checks, cheating always pays. The stable outcome is that both randomise, which is why every audit rate in this paper is a probability rather than a schedule.

A *mixed strategy* is exactly that randomisation — a probability distribution over what to do rather than a choice. Two players are in a *mixed equilibrium* when each is *indifferent* between its options given what the other is doing, since a player who strictly preferred one option would simply take it. Solving for indifference is how the critical rates below are derived, and it is the whole technique.

A player's behaviour is *incentive-compatible* when honesty is at least as profitable as the alternative — not when it is enforced or morally preferred. Everything this paper deters, it deters by making cheating cost more than it pays.

A *bond* is capital the judge posts before working and forfeits if caught cheating. It is the paper's central instrument, and it differs from a wage in a way that matters: a wage is paid for output and withheld for non-performance automatically, whereas a bond is forfeited only when a *detection event* occurs. That gap — between misbehaving and being caught misbehaving — is what the audit probability ρ and detection probability d measure, and it is why ρdB rather than B is the quantity a rational judge weighs. To *slash* a bond is to confiscate it on detection; the term is borrowed from proof-of-stake systems, where it means the same thing.

The two *settlement conventions* of §4 are the two natural answers to what happens to the corrupt gain itself when a cheat is caught. Under *keep-gain* the cheater keeps G and loses the bond; under *confiscation* the cheater loses both. They are not a modelling detail — they move the required audit budget by roughly 17% at the worked parameters, which is the paper's first practical finding.

A *bribe floor* is the least a briber must offer a judge to make corruption worth its while: exactly the judge's expected forfeiture $\beta = \rho dB$. Below that, no rational judge accepts.

Finally, *amortization* is the accounting sense of the word, not the algorithmic one: spreading a fixed cost over many uses so the per-use share falls. The paper's title claim is that a reputation *is* this operation applied to verification cost — the reason a long-trusted judge can be audited less is that its history has already paid for the confidence.

4 The stage game and the convention fork

A judge grades one outcome. Cheating yields a one-shot gain G. The platform audits with probability ρ at cost a per audit; an audit detects a cheat with probability d; detection slashes the judge's posted bond B. Under the *keep-gain* convention the caught cheat retains G and forfeits B; under the *confiscation* convention capture forfeits the gain as well.

Theorem 1 (Stage deterrence)**.** *A judge with one-shot corrupt gain G, facing audit probability ρ, detection probability d, and slashable bond B, is deterred iff $\rho dB \geq G$; the critical committed audit rate is*

$$\rho^\star = \frac{G}{dB} \qquad \textit{(keep-gain; Becker's condition).}$$

If capture also confiscates the gain, the cheat payoff is $G - \rho d(G+B)$ and the critical rate falls to

$$\rho_c^\star = \frac{G}{d(G+B)} < \rho^\star.$$

In the simultaneous (uncommitted) game, with a the audit cost and L the auditor's loss from an uncaught cheat, the mixed Nash equilibrium is $\rho^\star = G/(dB)$, $q^\star = a/(dL)$: the judge's indifference pins the audit rate, the auditor's indifference pins the cheat rate.

Why the same $\rho^\star$ twice. The formula is unchanged across the two game forms not by coincidence but because both solve the same judge-indifference equation: commitment picks the cheapest rate that deters, and simultaneous play lands an equilibrium mix exactly there.

Proof. Keep-gain: the cheat payoff is $G - \rho dB$, nonpositive iff $\rho \geq G/(dB)$. Confiscation: replace B by $G + B$ in the forfeiture. Mixed equilibrium: the judge mixes only if indifferent, $G - \rho dB = 0$; the auditor mixes only if indifferent between auditing $(-a - qL(1-d))$ and not $(-qL)$, which solves to $q^\star = a/(dL)$. Direct best-response checks confirm both indifferences numerically (§9). □

Numbers by hand. At the program's running parameters G=10, d=0.8, B=50: $\rho^\star = 10/(0.8{\cdot}50) = 0.25$ [verified] — audit one grade in four, and the cheat payoff is exactly +0.000000 at $\rho^\star$ [internal, `b2_tower.py`]. Under confiscation, $\rho^\star_c = 10/(0.8 \cdot 60) = 0.2083$ [verified]. The convention moves the required audit budget from 0.25 to 0.2083 — a 16.7% cut for free, *if and only if* your settlement rail actually claws back the gain. With a=1, L=40: $q^\star = 1/(0.8 \cdot 40) = 0.03125$ [verified]. Figure 2 plots the frontier $\rho dB = G$ in the (B, ρ) plane with this point marked. *Now you try:* double the bond to B=100 — what is $\rho^\star$? (0.125: a bigger bond lets you audit half as often; capital and vigilance are substitutes along $\rho dB = G$.)

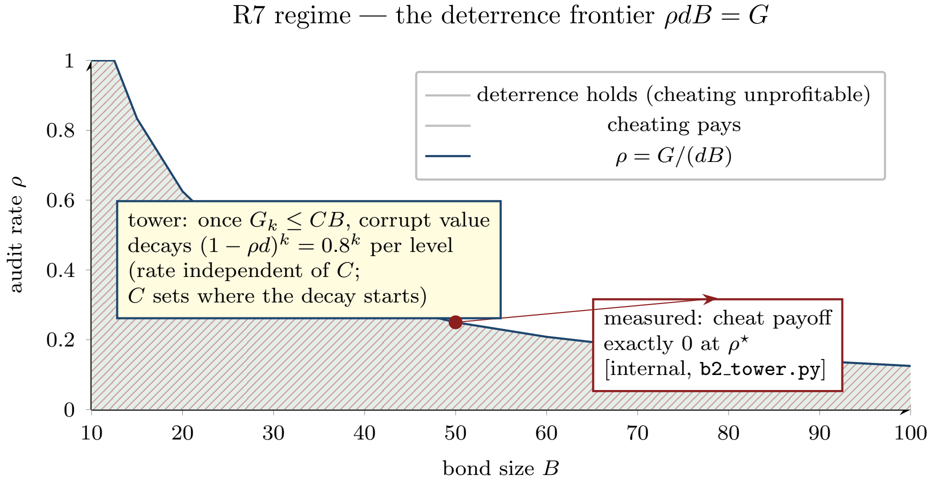


Figure 2: The regime diagram: the deterrence frontier $\rho dB = G$ in the (bond B, audit rate ρ) plane at G=10, d=0.8. Above the curve $\rho = G/(dB)$, cheating is unprofitable; below it, cheating pays. The marked point is the running example $(B, \rho^\star) = (50, 0.25)$, where the measured cheat payoff is exactly zero [internal, `b2_tower.py`]. Below the curve, the region is hatched rather than tinted so the two half-planes stay distinguishable in greyscale. The inset records the tower consequence of §5: below the bribery threshold $G_k \leq CB$, surviving corrupt value decays as $(1-\rho d)^k$ — a ×0.8-per-level collapse at these parameters, at every C. What C controls is not the rate but where the decay begins.

5 The tower: bribing sealed juries from C cliques

Two terms first, because both are hypotheses of the theorem below rather than hygiene recommendations. *Sealed* sampling means the briber must commit its money before learning which judge will handle the case; the entire clique multiplier vanishes without it, as the sealing discussion after the box shows. A *clique* means a pool of judges whose errors and loyalties are statistically and economically independent of another pool's — distinct vendors running distinct model families. The word is borrowed from social structure, not from graph theory, and it carries no adjacency-matrix meaning in this paper.

Who audits the auditors? Let level k+1 audit level-k auditors. An auditor at any level is itself a bonded judge, so Theorem 1 prices its honesty: the minimum bribe an auditor rationally accepts is its expected forfeiture $\beta = \rho dB$. The platform's one structural lever is *where auditors come from*: each audit is sampled, sealed, from a pool spanning C disjoint cliques — distinct principals, distinct model families — such that a corrupt agreement inside one clique does not reach the others. A briber protecting corrupt value G_k at level k faces a choice: bribe, or absorb the audit.

Theorem 2 (Tower contraction)**.** *Let level $k{+}1$ audit level-k auditors, each audit sampled sealed (the draw unknown to the briber) from a pool spanning C disjoint cliques, each auditor bonded at B with audit parameters (ρ, d) and bribe floor $\beta = \rho d B$. Then the briber's expected net value is affine in the number of cliques bought, $c \in \{0, \dots, C\}$: buying c still leaves the un-bought $(C{-}c)/C$ share of the sampling weight exposed to the same per-draw audit-and-detect risk ρd, so the payoff has no interior optimum and rational bribery is all-or-nothing — partial bribery is (weakly) dominated by one of the two corners. Bribing all C is profitable iff*

$$G_k\,\rho d \;>\; C\beta \quad\Longleftrightarrow\quad G_k \;>\; C\,B.$$

Below that threshold bribery stops and corrupt value decays geometrically:

$$G_{k+1} \;=\; (1-\rho d)\,G_k.$$

Consequently finite *bond capital certifies a tower deep enough to drive surviving corrupt value below any fixed unit, at B per level. If $G_0 \le CB$ the decay is geometric throughout and the certified depth is $\lceil \log G_0 / \log \frac{1}{1-\rho d} \rceil$. If $G_0 > CB$, a linear phase of $\lceil (G_0 - CB)/(C\rho d B) \rceil$ levels precedes it — during which the briber bleeds $C\beta$ per level — followed by $\lceil \log(CB)/\log \frac{1}{1-\rho d} \rceil$ geometric levels. Either way the depth is finite: logarithmic in the corrupt value that remains once bribery has stopped paying, plus that linear prefix if it was still paying at level 0. Raising C shortens or deletes the linear phase; it does not change the geometric rate $(1-\rho d)$, which is what makes the tower converge.*

Proof sketch. Buying c of the C cliques costs $c\beta$ up front (the seal forces payment before the draw is known) and retains value G_k if the draw lands on a bought clique (probability c/C) or $(1-\rho d)G_k$ if it lands on an honest one (probability $(C{-}c)/C$). Expected net value is

$$V(c) \;=\; G_k\Big(1 - \tfrac{C-c}{C}\rho d\Big) - c\beta \;=\; V(0) + c\Big(\tfrac{G_k \rho d}{C} - \beta\Big),$$

affine in c, so the maximum over $c \in \{0, \dots, C\}$ sits at a corner: bribery is all-or-nothing. Bribing beats not bribing iff $V(C) > V(0)$, i.e. $G_k\rho d > C\beta = C\rho d B$, i.e. iff $G_k > CB$; the audit bite ρd cancels, leaving a pure capital comparison between the value protected and the bonds that must be compensated. Once $G_k \le CB$ the geometric recursion takes over, and the level count to drive the surviving value below one unit from there is $\lceil \log \min(G_0, CB)/\log \frac{1}{1-\rho d} \rceil$, each level requiring one bond B; if $G_0 > CB$, the linear phase adds $\lceil (G_0 - CB)/(C\rho d B) \rceil$ levels ahead of it. $\square$

Why sealing is load-bearing, not hygienic. The factor C in the profitability threshold exists *only because the briber cannot see the draw.* If the sampling draw leaks — the briber learns *which* auditor will handle its case before the bribe is placed — it need buy exactly one auditor at price β, and bribery is profitable iff $G_k\rho d > \beta$, i.e. iff $G_k > B$: the clique multiplier vanishes and heterogeneity buys nothing. A leaked draw converts an eight-bench jury system into a single corruptible clerk. Sealed randomized selection (in deployment: VRF-selected honeypot audits, §8) is therefore part of the theorem's hypothesis, and auditing the *sampler* is part of the security boundary.

Numbers by hand. $G_0 = 400$, parameters as above, $\rho = \rho^\star = 0.25$, so $\beta = \rho d B = 10$ and $\rho d = 0.2$. At $C{=}1$: the profitability threshold is $G_k > 50$, so bribery still pays at first, and corruption bleeds only the bribe bill $C\beta = 10$ per level — $400 \to 390 \to 380 \to \cdots$ — for 35 levels, until G_k reaches $50 = CB$ and the geometric phase takes over for $\lceil \log 50/\log 1.25 \rceil =$ 18 more: 53 levels, $53 \times 50 = 2650$ of bond capital [verified, arithmetic; regenerates from b2_tower.py]. At C=8: the threshold is $G_k > 400$, not met even at level 0, so there is no linear phase at all and the collapse is geometric from the start at $(1-\rho d) = 0.8$ per level: $400 \to 320 \to 256 \to 204.8 \to \cdots$, reaching below one unit in $\lceil \log 400/\log 1.25 \rceil = 27$ levels, so total bond capital $27 \times 50 = 1350$ [verified, arithmetic].

What the clique multiplier actually buys. Set the two runs side by side and the multiplier is precisely a factor of two in certified depth and capital — 53 levels and 2650 against 27 and 1350 — not the difference between a tower that converges and one that does not. Both converge. Raising C *removes the linear phase*, because the profitability threshold CB rises past G_0; the geometric phase, and the rate $(1-\rho d)$ that drives it, are common to both. That is a real and budgetable saving, and it is the whole of the claim. *Now you try:* at C=2, what is the per-level bleed while bribery remains profitable, and where does the linear phase stop? ($C\beta = 20$ per level: $400 \to 380 \to 360 \to \cdots$, until $G_k \le CB = 100$ after 15 levels, where the decay turns geometric for a further 21 — 36 levels, 1800 of bond.)

Why the per-level condition suffices. $V(c)$ scores a *single* level's deviation, so $G_k > CB$ is an iff for one-level bribery, not for a bribery strategy. The distinction matters because buying level k+1 protects G_k only from that level's audit: level k+2 still threatens it, so a briber committed to the programme pays $C\beta$ at every level for as long as the tower runs. Scoring the strategy rather than the level makes bribery *less* attractive, not more, so the contraction is unaffected — at C=1 the all-bribe path spends $35 \times 10 = 350$ to shepherd $G_0 = 400$ down to 50 and then loses the 50 geometrically anyway, burning 87.5% of the protected value to arrive where it would have arrived regardless [verified, arithmetic]. The per-level condition is therefore the weaker and the safer of the two to state: it is what the affine argument establishes, and any multi-level bribery programme is dominated by it a fortiori.

6 Reputation is amortized verification

The stage game prices one grade. A platform judges T of them per judge lifetime, and flat auditing at $\rho^\star$ costs $a\rho^\star T = \Theta(T)$. But an old judge is not a newcomer: it has a verified history of length t, and with per-period reputation rent v its accumulated stake vt — the future income its record commands — is forfeit on capture. That stake is punishment capital the platform did not fund, and an incentive-compatible audit schedule may spend it.

Theorem 3 (Amortization). *Let a judge at history length t have reputation-at-stake vt, and let the audit schedule ρ_t be chosen so the one-shot deviation is weakly unprofitable at every t. Then:*

1. ***(Flat.)*** *The history-blind schedule $\rho_t \equiv \rho^\star = G/(dB)$ is incentive compatible and spends $a\rho^\star T = \Theta(T)$.*
2. ***(Model A: loss only if audited.)*** *If a cheat is penalized only when an audit catches it, and capture forfeits both bond and stake, the binding schedule is $\rho_t = G/\big(d(B+vt)\big)$, and lifetime spend is $a\sum_{t<T}\rho_t = \frac{aG}{dv}\ln\big(1+\frac{vT}{B}\big) + O(1) = \Theta(\log T)$.*
3. ***(Model B: independent revelation.)*** *If cheats additionally surface without audits at rate r per period (whistle-blowers, downstream failures, tombstoned provenance), destroying the stake vt when they do, the binding schedule is $\rho_t = \max\big(0, \frac{G-rvt}{dB}\big)$, which reaches zero at $t^\star = G/(rv)$; lifetime spend is bounded by the closed form*

$$a\sum_{t\ge 0}\rho_t \;\longrightarrow\; \frac{aGt^\star}{2dB} \;=\; \frac{aG^2}{2dBrv} \;=\; O(1).$$

Under each schedule the deviation value is ≤ 0 pointwise at every t: reputation does not *weaken deterrence, it* funds *it.*

Proof. (2) The deviation value at t is $G-\rho_t d(B+vt)$; setting it to zero gives the schedule, and the spend is a harmonic-type sum: $a\sum_t G/(d(B+vt)) \approx \frac{aG}{dv}\ln(1+vT/B)$. (3) The deviation value is $G-\rho_t dB - rvt$ (audit slash plus the expected independent loss); the binding schedule is linear and hits zero at $t^\star = G/(rv)$, so the spend is the area of a triangle with base $t^\star$ and height $a\rho^\star = aG/(dB)$: $\frac{1}{2}\cdot\frac{G}{rv}\cdot\frac{aG}{dB} = \frac{aG^2}{2dBrv}$, independent of T. □

Numbers by hand — with the caveat a referee would otherwise find. Parameters G=10, d=0.8, B=50, a=1, v=0.6, r=0.05, horizon T=200. Flat spend: $1\cdot 0.25\cdot 200 = 50.00$ [verified]. Model A: 25.58 measured against closed form $\frac{aG}{dv}\ln(1+vT/B) = 25.50$ [internal, `b2_tower.py` / verified]. Model B: measured spend 35.08 against the $O(1)$ limit $aG^2/(2dBrv) = 41.67$ [internal/verified]. **Note that** $35.08 \neq 41.67$**, and the discrepancy is not error:** at these parameters $t^\star = G/(rv) = 333 > T = 200$, so the audit schedule has not yet reached zero within the measured horizon — *the* $O(1)$ *curve had not saturated at* T=200. The measured 35.08 is the partial triangle; the closed form is its full area, attained only for $T \geq t^\star = 333$. Run the horizon past $t^\star$ and the cumulative spend flattens and stays there. Figure 3 shows both halves of this: panel A is the measured range, where all three curves are near-linear and the $O(1)$ curve is in fact the second-steepest; panel B runs the same three schedules to T=1500, where flat reaches 375.00, Model A 61.46 and is still climbing, and Model B stops dead at 41.79 from $t^\star$ onward and never spends again [verified, `paper3_amortization_figures.py`]. (The plateau is 41.79 rather than 41.67 because the closed form is the area of the continuous triangle while the schedule is summed period by period; the excess is exactly half the first term, $\rho^\star/2 = 0.125$.) That is why the pre-asymptotic gap at T=200 is a check on the closed form rather than a strike against it: read against $t^\star$, not against the limit, 35.08 is precisely the partial area the theorem predicts. The incentive checks pass pointwise: maximum deviation value $+1.78\times 10^{-15}$ under both schedules — machine-epsilon zeros, i.e. ≤ 0 up to float epsilon, exactly the binding-constraint signature [internal]. *Now you try:* at what T do Model A and flat differ by a factor of 4? (Solve $\frac{aG}{dv}\ln(1+vT/B) = \frac{1}{4}a\rho^\star T$; at these parameters $T \approx 779$ — the log curve's lead widens without bound.)

The fork is empirical. Whether a deployment lives in Model A or Model B is a question about the world, not the model: *do cheats surface without audits?* If bad grades eventually reveal themselves — failed downstream builds, provenance tombstones, user reports — then $r > 0$ and lifetime verification spend per honest judge is a constant. If a cheat undetected at audit time is silent forever, only the $\Theta(\log T)$ schedule is safe. The deployment telemetry that estimates r (the rate at which sanctioned outcomes were first flagged by non-audit channels) is therefore part of this paper's measurement obligation, not an implementation detail.

7 The discount-factor connection: claim signaling, mechanized

The program's repeated-game track prices the same deterrence with a different currency. In the corrected claim-signaling stage game — payoffs $(3,3)$ mutual-truth, $(0,4)/(4,0)$ sucker/defector, $(1,1)$ mutual-claim — the one-shot deviation gain is $g=1$ and the per-round punishment loss is 2. By the one-shot deviation principle, a grim trigger is subgame-perfect — cooperation survives every history, not just the equilibrium path — iff $1 \leq 2\,\delta/(1-\delta)$, i.e. $\delta \geq 1/3$; the gentler three-round graduated trigger is subgame-perfect iff $1 \leq 2(\delta+\delta^2+\delta^3)$, whose threshold is the unique real root in $(0,1)$ of $2\delta^3+2\delta^2+2\delta-1=0$, numerically $\delta^\star = 0.3425080314$. These are not folklore numbers: they are CI-mechanized in the repository's `proofs/economics/` directory — `delta-threshold.z3` proves existence, uniqueness in $(0,1)$, and location in $[0.34, 0.35]$ of the root by real-algebraic decision procedure; `claim_signaling.tla` model-checks the discounted one-shot-deviation invariant under both TLC and Apalache; and `sweep-delta.sh` confirms the empirical crossover ($\delta = 0.35$ on the integer grid) matches the closed form. Every claim in this paragraph is [verified] by that CI suite.

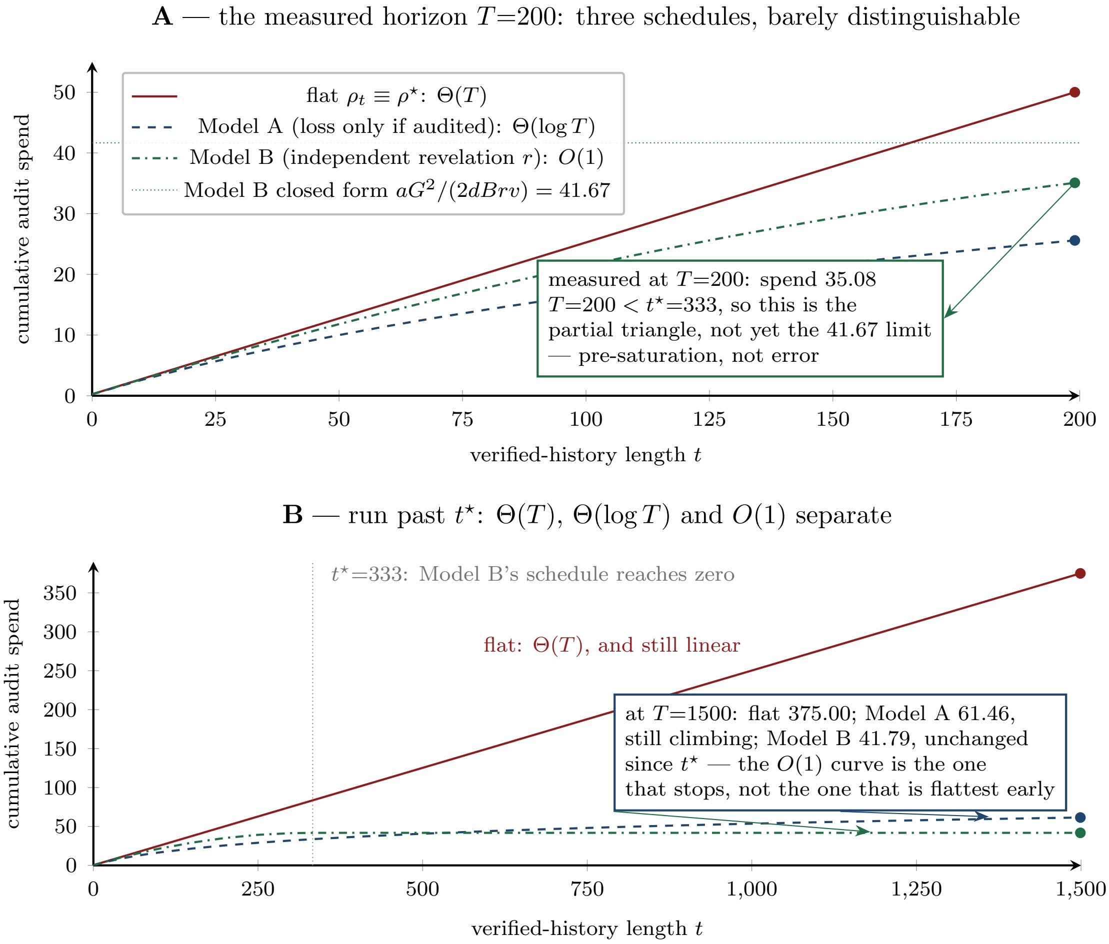


Figure 3: Cumulative audit spend against verified-history length t under the three incentive-compatible schedules of Theorem 3, at the running parameters. **A:** the horizon the paper measures, T=200. All three curves are near-linear here and Model B — the $O(1)$ schedule — is the *second-steepest*: on this range the asymptotics are not visible, and the measured 35.08 sits below the closed-form 41.67 because saturation requires $t^\star = G/(rv) = 333$. That gap is pre-asymptotic, not error. **B:** the same three schedules run to T=1500, past $t^\star$. Now the growth classes separate on the page: flat reaches 375.00 and keeps climbing linearly, Model A reaches 61.46 and is still climbing (logarithmically), and Model B stops dead at 41.79 from $t^\star$ onward and never spends again. Model A's cumulative spend crosses *above* Model B's at t=532: past that point the $O(1)$ schedule is permanently the cheaper of the two, which is the reversal panel A cannot show. The plateau 41.79 exceeds the continuous closed form 41.67 by $0.125 = \rho^\star/2$, the expected discrete-sum overshoot on a linear ramp. Line styles differ (solid / dashed / dash-dot) so the three series survive greyscale printing [verified: `paper3_amortization_figures.py`, seed 20260816; cross-checks against `b2_tower.py`].

The connection to Theorem 3 is exact, not analogical. In Kreps–Wilson [3] and Fudenberg–Levine [4], deterrence is funded by *continuation value*: the discounted future surplus a deviator forfeits. In the inspection game it is funded by *bond plus stake*: $d(B + vt)$ at audit, rvt by revelation. Model A's schedule $\rho_t = G/(d(B + vt))$ is precisely the statement that as endogenous continuation value vt accumulates, exogenous verification ρ_t can retreat while the total deterrent stays pinned at G — the bonded game continuously deforms into the repeated game. The discount thresholds $\delta \geq 1/3$ and $\delta^\star = 0.3425$ are the limiting case $B = 0$, $\rho = 0$: all deterrence carried by the future. Reputation is amortized verification in both formalisms; the inspection game just itemizes the invoice.

8 Conjecture III.11.1, parameterized

The source volume's central open question — Conjecture III.11.1, the grading-oracle problem — asked whether a recursive rate-the-raters scheme can terminate. Two external reviews proposed answers that seemed to compete: one mandated a heterogeneous judge mix (distinct model families, to break correlated mode collapse) with VRF-selected honeypot tasks so a judge never knows when it is audited; the other derived the local condition $\rho dB > G$ and insisted a recursive tower needs a per-level contraction plus a grounded termination. Neither had the theorem. Fused, they *are* it:

- *Model heterogeneity supplies the cliques.* The disjointness that Theorem 2's C counts is statistical and economic independence of judge pools — exactly what distinct principals running distinct model families provide, and what a homogeneous pool (one vendor, one model, C=1) does not. The cost of C=1 is a linear bribery phase before the geometric one, roughly doubling certified depth and bond capital at the worked parameters (53 levels and 2650 against 27 and 1350); heterogeneity buys the removal of that phase, not the difference between a tower that converges and one that does not (§5).
- *VRF honeypots implement the sealed audit.* A verifiable random function [6] selecting honeypot tasks at rate $\rho^\star$ is precisely the sealed sampling the contraction hypothesis requires: the draw is unpredictable to the briber (sealing) yet publicly auditable after the fact (so the sampler itself cannot be quietly corrupted).
- *The termination is named, not wished.* The tower roots in an exogenously honest anchor — the operator at n=1, a bonded arbitration market at scale — and Theorem 2 prices the depth: 27 levels and bond capital 1350 at the running parameters [verified, arithmetic]. Concretely, this is what the judge ceremony of the reputation chapter is buying: the re-audit sampling rate is $\rho^\star$, the "distinct architectural weights" quorum is the C in CB, the VRF injection is the seal, and the depth-and-capital line item ($27 \times B$) is a number a platform can put in a budget rather than an assumption it has to hope holds.

The conjecture thereby becomes a *parameterized theorem* — deterrence at $\rho^\star = G/(dB)$ per level, contraction $(1 - \rho d)$ under sealed C-clique sampling with the bribery cutoff $G_k \leq CB$, amortized lifetime cost by Theorem 3 — plus a *measurement obligation*: does the deployed judge pool actually satisfy sealed multi-clique sampling, and what are d, r, and the cross-clique collusion correlation? Those are numbers to estimate from audit telemetry, not assumptions to wave at.

9 Reproducibility

Every number in this paper regenerates deterministically from one self-contained script, `skills/harbor-result` (numpy + matplotlib; program seed convention 20260816, though the computation is deterministic). Its output at the running parameters, verbatim:

```
rho* (keep-gain)     = G/(dB)      = 0.2500
rho* (confiscation)  = G/(d(G+B))  = 0.2083
q*   (mixed NE)      = a/(dL)      = 0.0312
judge indifferent at rho*:  cheat payoff = +0.000000
C=1:  400.0  390.0  380.0  370.0  360.0  350.0  340.0  330.0  320.0
C=8:  400.0  320.0  256.0  204.8  163.8  131.1  104.9   83.9   67.1
levels to drive corrupt value below 1 unit (geometric regime):  27;
total bond capital = B per level * levels = 1350
flat:       50.00   (Theta(T))
model A:    25.58   (Theta(log T); closed form 25.50)
model B:    35.08   (O(1); closed form aG^2/(2dBrv) = 41.67; hits 0 at t*=333)
IC check A: max deviation value = +1.78e-15 (<=0)
IC check B: max deviation value = +1.78e-15 (<=0)
```

The script also verifies the mixed equilibrium by direct best-response check (auditor indifference $-1.25 = -1.25$) and renders the three-panel summary figure. The claim-signaling thresholds of §7 regenerate independently via the CI suite in `proofs/economics/` (Z3, TLC, Apalache, and the δ-sweep).

10 Honest boundaries

What this paper does not say.

- *The headline is convention-dependent.* $\rho^\star = 0.25$ under keep-gain, 0.2083 under confiscation — a ~17% swing from the settlement rule alone. Quote neither without stating which convention your slashing contract implements; a platform whose escrow claws back the gain but which budgets audits at $G/(dB)$ is over-auditing, and one that budgets at $G/(d(G+B))$ without clawback is under-deterring.
- *Sealed sampling is a hypothesis, not a garnish.* A briber who learns the audit draw before placing bribes faces threshold $G_k > B$, not $G_k > CB$: the clique math collapses to C=1 (§5). The sampler — and its VRF, and whoever operates it — is inside the trusted computing base of Theorem 2.
- *The tower needs an honest root.* Contraction shrinks corruption level over level; something exogenously honest must sit at the top — the operator in a small deployment, a bonded arbitration market at scale. The theorem prices the depth to that root; it does not conjure the root.
- *d, r, v, and cross-clique correlation are measured quantities, not assumptions.* Detection probability comes from audit outcomes; the revelation rate r from non-audit provenance of sanctions; cross-clique collusion correlation from judge-pair agreement statistics. "Disjoint cliques" is falsifiable and must be monitored: two nominally rival benches fine-tuned from the same base model may fail disjointness in exactly the correlated-error cases that matter.
- *The $\Theta(\log T)$-vs-$O(1)$ fork is empirical.* Model B's constant lifetime spend holds only if cheats surface without audits ($r > 0$). And at the session parameters the $O(1)$ claim is demonstrated pre-asymptotically: $t^\star = 333 > T = 200$, so the measured spend 35.08 sits below, not at, the 41.67 limit (§6).
- *One-shot gains, risk-neutral bribers, stationary parameters.* G is per-grade; an adversary who can correlate many small cheats below audit granularity, or who values capture asymmetrically, needs the repeated-game extension of §7 with the stake re-derived. The

machine-epsilon IC residuals ($+1.78 \times 10^{-15}$) are zeros of a float computation, stated as "≤ 0 up to float epsilon," not as exact algebra.

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